AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 23 · STUDENT LESSON

# Training a small image model

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

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## Before you start

Why could pooling all four pixels of a 2 by 2 image hide whether the bright pixels are on the left or right?

Plan time to read, investigate and explain. Keep predictions separate from observations.

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STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Trace a complete image model with fixed features and one trained weight. Try the recall before opening notes.

### Your learning target

I can trace a complete image model with fixed features and one trained weight.

I can select a **checkpoint** using a declared validation rule.

I can evaluate a frozen model against a **baseline** and limit the conclusion.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Pooling and spatial information. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Initialisation and reproducibility. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Trace a complete image model with fixed features and one trained weight

Select a **checkpoint** using a declared validation rule

Evaluate a frozen model against a **baseline** and limit the conclusion

### Retrieve what you already know

> Why could pooling all four pixels of a 2 by 2 image hide whether the bright pixels are on the left or right?

**My starting explanation**

### This week's end goal

By the end, I can explain **training** a small image model using a traced example and a stated limitation.

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# Notice the evidence

> Investigate the supplied Training a small image model case. Separate observations, given facts and assumptions before choosing an explanation.

Use fictional 2 by 2 binary images, stored as two rows. Predict class 1 when the left column sum exceeds the right column sum; otherwise predict class 0. Equal sums belong to class 0.

This known arithmetic task makes every step checkable; it does not establish real-world image recognition.

| Split and ID | Image rows | Class | Difference d |
|---|---|---|---|
| Train A | [1, 0] and [0, 0] | 1 | 1 |
| Train B | [1, 0] and [1, 0] | 1 | 2 |
| Train C | [0, 1] and [0, 0] | 0 | -1 |
| Train D | [0, 1] and [0, 1] | 0 | -2 |
| Validation E | [0, 0] and [1, 0] | 1 | 1 |
| Validation F | [0, 0] and [0, 1] | 0 | -1 |

Apply the fixed kernel [[1, -1], [1, -1]] to the entire image to obtain d. Its negative gives -d. ReLU produces h1 = max(0, d) and h2 = max(0, -d). The **output** score is a(h1 - h2) = ad.

Only the shared scalar a is trained; both kernels stay frozen and there is no bias or pooling.

For squared-error **training**, map class 1 to target t = 1 and class 0 to t = -1. Predict class 1 only when score > 0; otherwise class 0. The score is not a probability.

Keep the teacher-held final cases unseen until the **checkpoint** is selected and recorded.

**List the given facts and one fact that is not supplied.**

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STUDENT LESSON · 3 OF 18

## See how it works

> Follow the mechanism for Training a small image model. Connect each labelled input, deciding step and result.

Start a = 0. Use all four **training** examples in each update, ordinary mean squared error and learning rate 0.1. Save **checkpoint**s at updates 0, 1 and 2. The **training** loss is 2.5a squared - 3a + 1, with gradient 5a - 3.

Subtract learning rate times gradient; this gives a = 0.3 then a = 0.45.

```
def feature(image):
    return image[0][0] + image[1][0] - image[0][1] - image[1][1]

train = [([[1, 0], [0, 0]], 1),
         ([[1, 0], [1, 0]], 1),
         ([[0, 1], [0, 0]], -1),
         ([[0, 1], [0, 1]], -1)]
valid = [([[0, 0], [1, 0]], 1),
         ([[0, 0], [0, 1]], -1)]

def mse(a, rows):
    return sum((a * feature(x) - t)**2 for x, t in rows) / len(rows)

a, checkpoints = 0.0, [0.0]
for _ in range(2):
    gradient = sum(2 * (a * feature(x) - t) * feature(x)
                   for x, t in train) / len(train)
    a -= 0.1 * gradient
    checkpoints.append(a)
```

```
for step, value in enumerate(checkpoints):
    print(step, round(value, 6), round(mse(value, train), 6),
          round(mse(value, valid), 6))
```

The code assumes valid 2 by 2 numeric images and nonempty labelled lists. It prints update, weight, **training** MSE and validation MSE. Validation examples are scored but never used in the gradient.

A single-channel filter spanning the whole image produces one value here, so no further spatial pooling is needed.

### Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

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STUDENT LESSON · 4 OF 18

## Compare the cases

> Compare the supplied cases for Training a small image model. Identify the detail that changes the conclusion.

### Compare a familiar case with a changed case

Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

### Change one thing in the pictured case

#### Change one thing

Change: Double the two-update budget while keeping the same **training** rule.

#### Predicted result

**Prediction**: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

#### Why the result follows

Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

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STUDENT LESSON · 5 OF 18

## Words that unlock the idea

> Use the vocabulary of Training a small image model to explain a concrete decision, rather than only memorising definitions.

| Word | Meaning |
| --- | --- |
| Frozen feature | A feature computation whose parameters are kept unchanged during training |
| Checkpoint | Saved model parameters at a particular training stage |
| Selection metric | The quantity used under a declared rule to choose a model |
| Baseline | A simple comparison method evaluated on the same cases |

**Use two terms to explain a step in this week's example.**

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# Follow a worked example

Follow the worked trace for Training a small image model, predict a step, then compare with its explanation.

### Compare the saved checkpoints

| UPDATE AND WEIGHT | TRAINING MSE | VALIDATION MSE |
|---|---|---|
| 0 with a 0 | 1 | 1 |
| 1 with a 0.3 | 0.325 | 0.49 |
| 2 with a 0.45 | 0.15625 | 0.3025 |

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

**At a = 0.3 and a = 0.45, compute the validation scores for E and F and their predicted classes. Why can validation accuracy stay the same while validation MSE changes?**

### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

### A pictorial worked case: Training a small image model

Train one tiny image score with a fixed budget and a separately reserved check.

### The facts we will use

#### Supplied facts

Toy brightness difference d=left−right; score=a×d. **Training** pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates.

#### Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

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# Training a small image model

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.

### 1. First update

Gradient 2(2a-2) x 2=-8; new a = 0.5, training loss 1.

### 2. Second update

Gradient at 0.5 is-4; new a = 0.75, training loss 0.25.

### 3. Validation score

At reserved d1, target 1: prediction 0.75, squared error 0.0625.

**CHANGE ONE THING**

Double the two-update budget while keeping the same training rule.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. First update | Gradient 2(2a-2)×2=-8; new a = 0.5, training loss 1. |
| 2. Second update | Gradient at 0.5 is-4; new a = 0.75, training loss 0.25. |
| 3. Validation score | At reserved d1, target 1: prediction 0.75, squared error 0.0625. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

### Why the result follows

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

### What this example does not establish

#### Keep the stated limit

This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

#### Keep the stated limit

Read the labelled picture: Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

#### Keep the stated limit

Change one thing: Double the two-update budget while keeping the same **training** rule. Predict the result before checking the explanation. What remains the same?

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STUDENT LESSON · 7 OF 18

## Find and repair a mistake

> Find the unsupported step in the Training a small image model claim and explain a repair using the supplied evidence.

#### Check the claim against the facts

Save a = 0.45, both fixed kernels, the **input** schema, target mapping, strict decision threshold, split IDs, learning rate, update budget and selection rule.

The **baseline** always predicts class 0: **training** classes are tied, and the declared tie rule chooses 0. It gets 2 of 4 **training** cases and 1 of 2 validation cases correct. The selected model gets all of those cases correct.

### Check the claim against the facts

The teacher now supplies the three reserved final images. Record each score and **prediction** before comparing with the supplied labels.

Report correct count and denominator for both the frozen model and the same constant **baseline**. Do not tune the threshold or retrain after viewing final errors and still call that same set unseen.

### Check the claim against the facts

Because the images and target rule were constructed for this small exercise, even perfect results support only a narrow claim. Validation images differ from **training** pixels but reuse differences 1 and -1.

This is a deliberately simple check of a procedure, not **evidence** of broad generalisation.

### Locate the earliest unsupported step

Select among the three saved lesson **checkpoint**s using validation MSE, with earliest update breaking a tie. Is selecting by **training** accuracy the same declared procedure?

**State one repair and the checks you would rerun.**

# Practise together

> Practise Training a small image model with support, then explain the deciding step in your own words.

For Train B, compute d, h1, h2 and the score at a = 0.45. Give the class **prediction** and squared error against its signed target.

**At a = 0.3 and a = 0.45, compute the validation scores for E and F and their predicted classes. Why can validation accuracy stay the same while validation MSE changes?**

**Explain your prediction to a partner before comparing results.**

### Practise with less help: Check both updates

Use workbook W2 for the connected example "**Training** a small image model". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
|---|---|---|
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for **Training** a small image model. Proposed explanation: “All parameters in a trained model must have been learned”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> **My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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## Run the investigation

> Investigate Training a small image model. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

### Predict, investigate and preserve the evidence

Starting at a = 0, use gradient 5a - 3 and rate 0.1 for two updates. Give each gradient, new weight and **training** MSE.

```
def feature(image):
    return image[0][0] + image[1][0] - image[0][1] - image[1][1]

train = [([[1, 0], [0, 0]], 1),
         ([[1, 0], [1, 0]], 1),
         ([[0, 1], [0, 0]], -1),
         ([[0, 1], [0, 1]], -1)]
valid = [([[0, 0], [1, 0]], 1),
         ([[0, 0], [0, 1]], -1)]

def mse(a, rows):
    return sum((a * feature(x) - t)**2 for x, t in rows) / len(rows)

a, checkpoints = 0.0, [0.0]
for _ in range(2):
    gradient = sum(2 * (a * feature(x) - t) * feature(x)
                   for x, t in train) / len(train)
    a -= 0.1 * gradient
    checkpoints.append(a)
for step, value in enumerate(checkpoints):
    print(step, round(value, 6), round(mse(value, train), 6),
          round(mse(value, valid), 6))
```

| Record | Your evidence |
| --- | --- |
| Given input and fixed rule | |
| Prediction before checking | |
| Actual result, or “not run” | |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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STUDENT LESSON · 10 OF 18

## Build your understanding

> Move from recall to tracing, diagnosis and transfer in Training a small image model. Explain why each step is justified.

**Explain → apply → challenge**

For Train B, compute d, h1, h2 and the score at a = 0.45. Give the class **prediction** and squared error against its signed target.

> Select among the three saved lesson **checkpoint**s using validation MSE, with earliest update breaking a tie. Is selecting by **training** accuracy the same declared procedure?

After inspecting the final cases, a team changes the decision threshold and reports the best result as an unseen test score. Explain the problem and list two things the original report should preserve.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

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STUDENT LESSON · 11 OF 18

## Connect your project

> Apply this week's idea to Image-pattern Network Investigation. Keep an annotated decision and the evidence supporting it.

Add a labelled **training** a small image model **evidence** sheet to Image-pattern Network Investigation. Show how you can trace a complete image model with fixed features and one trained weight.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
| --- | --- |
| Component and version | |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

### Evidence for your project

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

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STUDENT LESSON · 12 OF 18

## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain

practice; use the separately named independent case before feedback.

**Training** adjusted one weight in a model with hand-chosen features. It did not discover the left-versus-right kernel.

The arithmetic rule itself would solve the declared task exactly; the trained model helps us study **fitting** and evaluation, not demonstrate that learning is necessary here.

**A teammate claims the model learned its own image features and that choosing the best score on the final test would improve the experiment. Correct both claims and name the evidence needed in the final report.**

Report the fixed and trained components separately, selected **checkpoint**, per-case final **prediction**s, **baseline** results and limitations.

If a final case fails, preserve that failure and design a new experiment with new reserved evaluation data. Next week examines augmentation and leakage between related images.

After inspecting the final cases, a team changes the decision threshold and reports the best result as an unseen test score. Explain the problem and list two things the original report should preserve.

**My limited claim and a useful next check**

# Study a complete case

> Revisit the complete Training a small image model case for supported practice. Repeated examples do not become fresh assessments.

### Complete evidence record

#### Compare the saved checkpoints

| UPDATE AND WEIGHT | TRAINING MSE | VALIDATION MSE |
|---|---|---|
| 0 with a 0 | 1 | 1 |
| 1 with a 0.3 | 0.325 | 0.49 |
| 2 with a 0.45 | 0.15625 | 0.3025 |

Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

> **My complete trace of the worked case**

# Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Choose a checkpoint independently

A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Attempt this without hints

A separate model scores bd. Its **training** examples have d = 1 and -1 with signed targets 1 and -1. Start b = 0; gradient is 2(b - 1), rate 0.25 and budget two updates. Validation differences are 2 and -2 with targets 1 and -1.

Calculate all three weights and their validation MSEs, then select by lowest validation MSE.

**My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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## Your lesson route

> Use this map to navigate Training a small image model, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Train one tiny image score with a fixed budget and a separately reserved check.

| By the end, I can | How I will show it |
| --- | --- |
| Trace a complete image model with fixed features and one trained weight | Explain the deciding step, show a trace or test, and state one limit. |
| Select a checkpoint using a declared validation rule | Explain the deciding step, show a trace or test, and state one limit. |
| Evaluate a frozen model against a baseline and limit the conclusion | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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## Finish and return

> Explain Training a small image model without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Trace a complete image model with fixed features and one trained weight

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

## Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Train one tiny image score with a fixed budget and a separately reserved check.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

1. Validation score

2. First update

3. Second update

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Validation score At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

### Practice 2: explain the deciding evidence

#### Changed case · predict first

Changed-case practice: Double the two-update budget while keeping the same **training** rule. Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. Next a values 0.875 and 0.9375; **training** loss falls to 0.015625.

**My selected conclusion, deciding fact and explanation**

#### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Each update halves the distance to the **training** optimum 1 in this toy system.

### Repair a missing building block

Review Pooling and spatial information (ai-5/22). Goal: Compute max and mean pooling under an explicit window rule. Short practice: Can the max map reveal where the 4 originally appeared?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Image tensors and channels (ai-5/20). Goal: Read a pixel using declared row column and channel axes. Short practice: Where is the bottom green pixel in the first batched image?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Initialisation and reproducibility (ai-5/17). Goal: Trace how starting values affect a fixed **training** budget. Short practice: Does the matching final loss mean the parameter values are equal?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

### Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

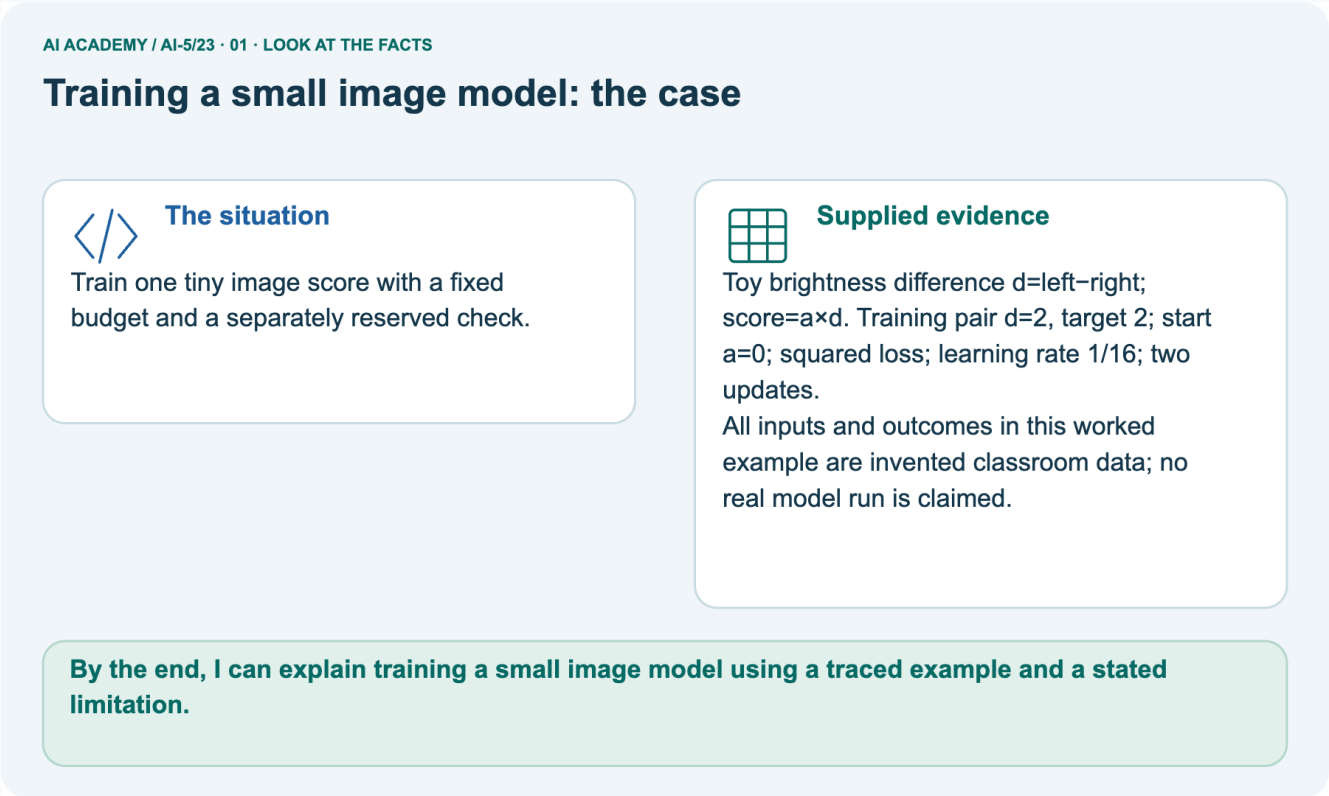
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STUDENT LESSON · 18 OF 18

# Visual concept journey

> Explain each Training a small image model graphic using its labels, deciding evidence and relevant limitation.

### 01. Training a small image model: the case



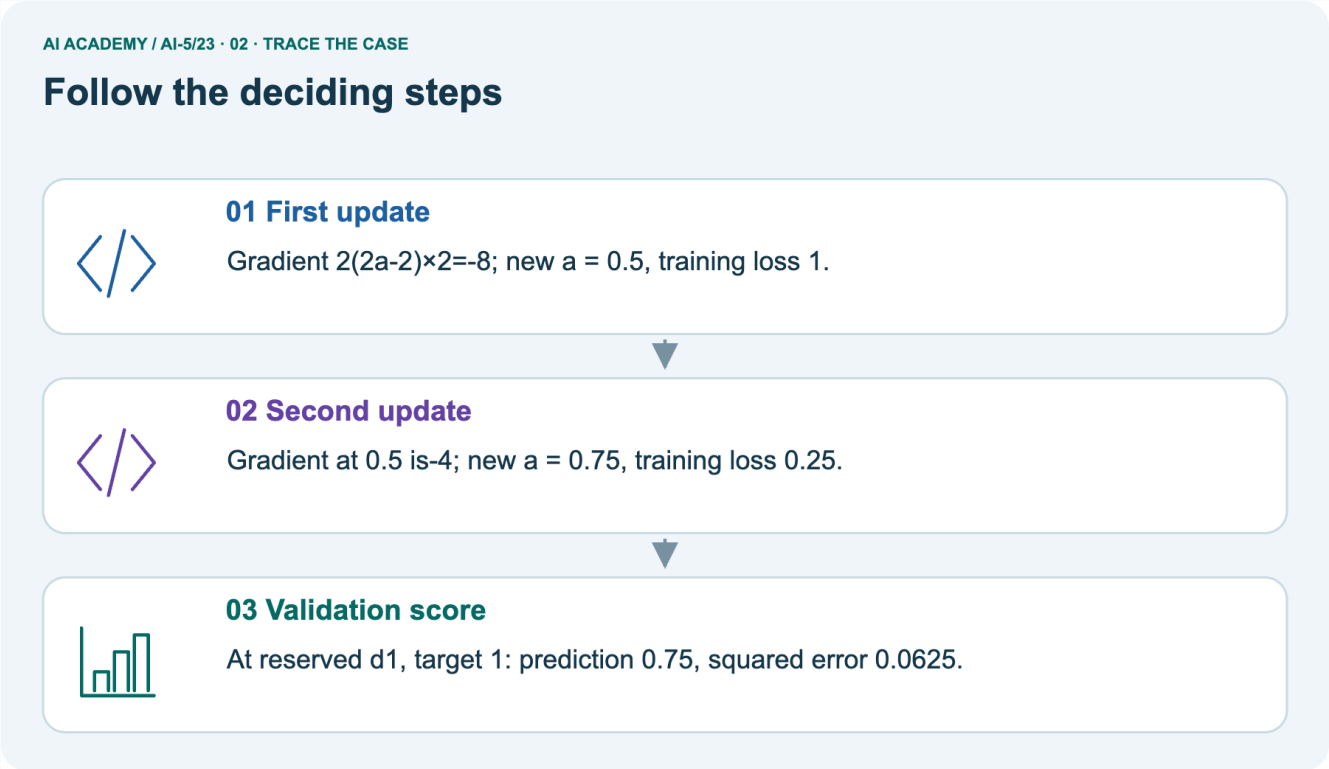

Follow the labelled picture with the matching explanation. Open the vector diagram

Train one tiny image score with a fixed budget and a separately reserved check. Supplied facts: Toy brightness difference d=left−right; score=a×d.

**Training** pair d=2, target 2; start a=0; squared loss; learning rate 1/16; two updates. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

### 02. Follow the deciding steps




Follow the labelled picture with the matching explanation. Open the vector diagram

First update: Gradient 2(2a-2)×2=-8; new a = 0.5, **training** loss 1. → Second update: Gradient at 0.5 is-4; new a = 0.75, **training** loss 0.25. → Validation score: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625.

> Point to each numbered stage and explain which recorded **input** it uses.

## 03. Compare the saved checkpoints

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### Compare the saved checkpoints

| Update and weight | Training MSE | Validation MSE |
|---|---|---|
| 0 with a 0 | 1 | 1 |
| 1 with a 0.3 | 0.325 | 0.49 |
| 2 with a 0.45 | 0.15625 | 0.3025 |

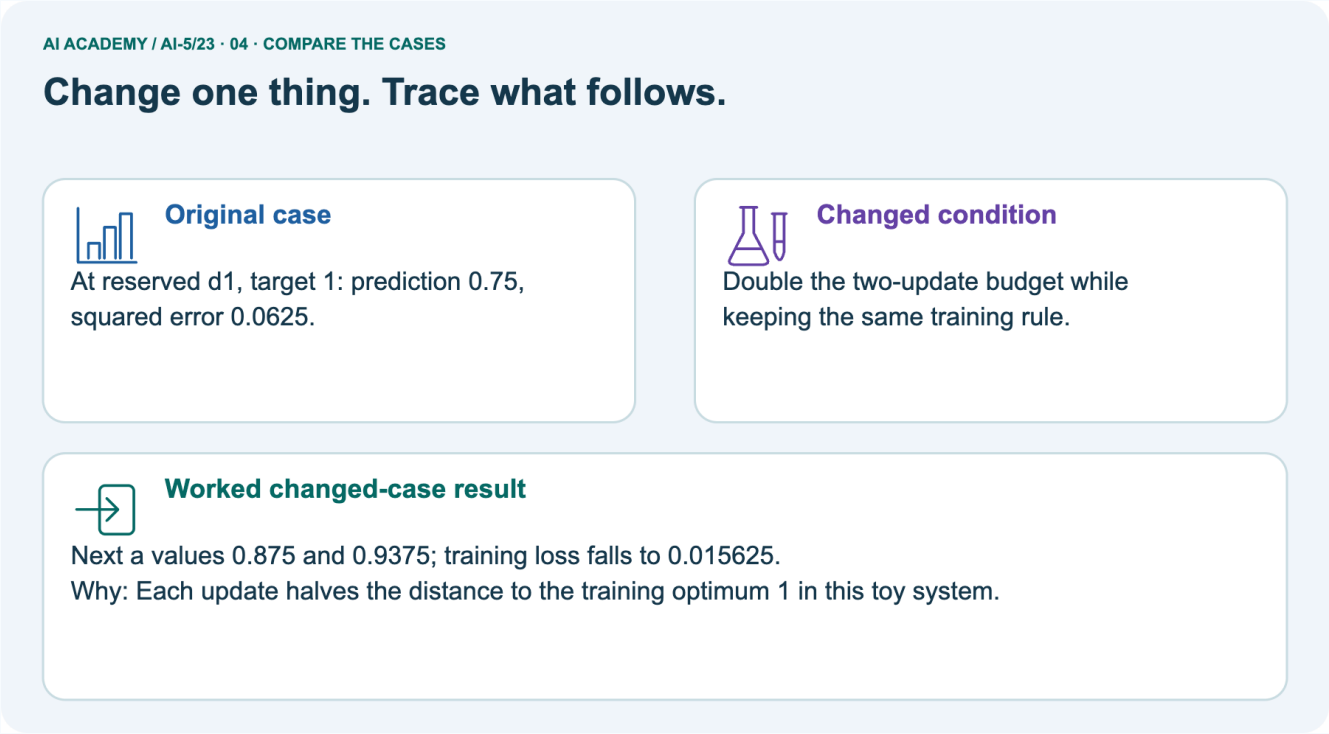
Before training, declare selection by lowest validation MSE among these three checkpoints, with the earliest checkpoint winning an exact tie. Checkpoint 2 is selected. Training loss alone does not define the selection rule.

Follow the labelled picture with the matching explanation. Open the vector diagram

Compare the saved **checkpoint**s. 0 with a 0 | 1 | 1. 1 with a 0.3 | 0.325 | 0.49. 2 with a 0.45 | 0.15625 | 0.3025

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



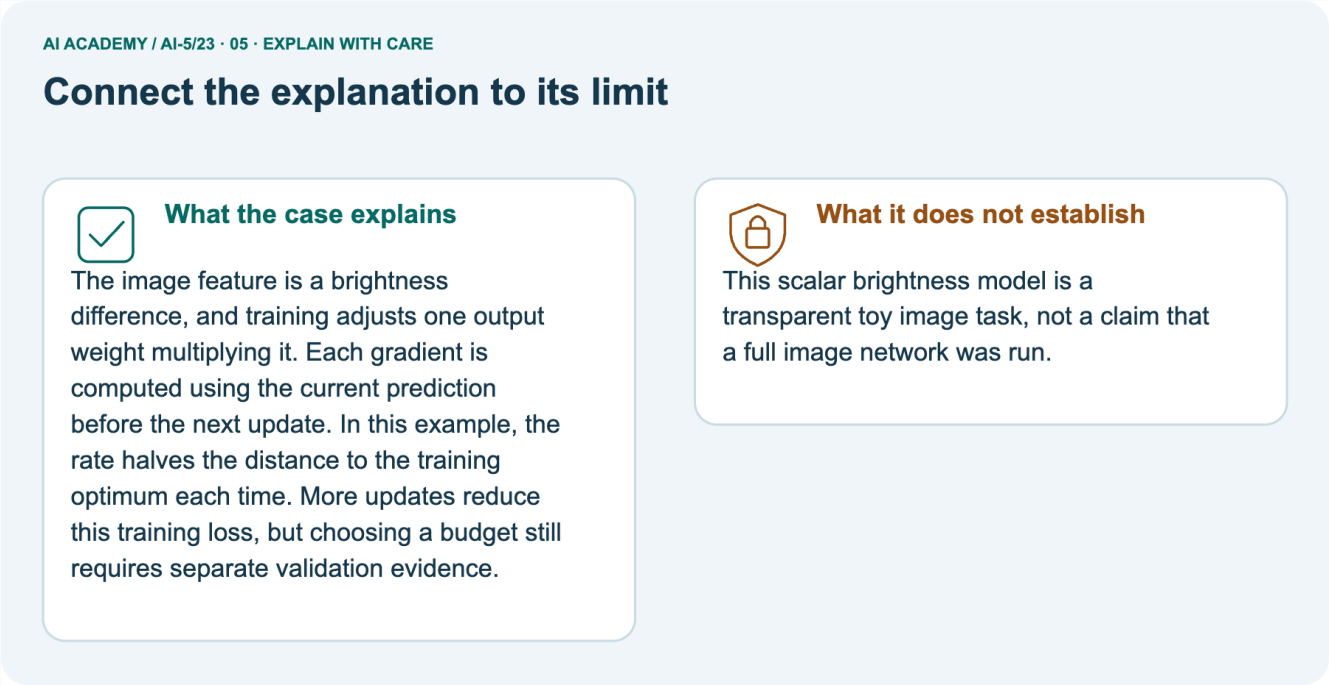

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: At reserved d1, target 1: **prediction** 0.75, squared error 0.0625. Changed condition: Double the two-update budget while keeping the same **training** rule.

Expected result: Next a values 0.875 and 0.9375; **training** loss falls to 0.015625. Reason: Each update halves the distance to the **training** optimum 1 in this toy system.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

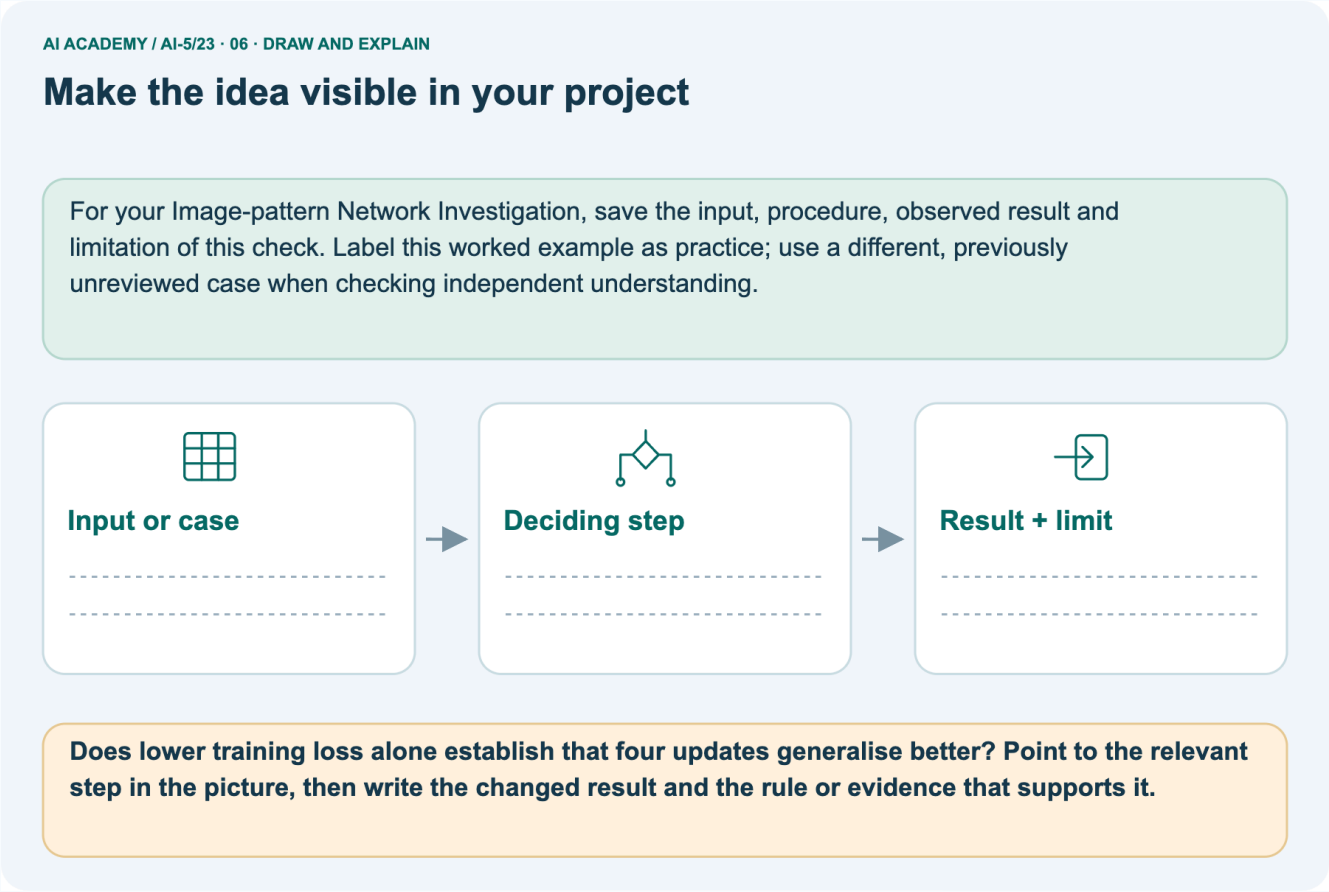
The image feature is a brightness difference, and **training** adjusts one **output** weight multiplying it. Each gradient is computed using the current **prediction** before the next update.

In this example, the rate halves the distance to the **training** optimum each time. More updates reduce this **training** loss, but choosing a budget still requires separate validation **evidence**.

Limit: This scalar brightness model is a transparent toy image task, not a claim that a full image network was run.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does lower **training** loss alone establish that four updates generalise better? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.